tp

Developer Guide

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Quick Start

Follow the steps below to set up and run the application:

Prerequisites

- Ensure that you have **Java 17** installed on your system.
- You can verify your Java version by running:

  ```
  java -version
  ```

- If the version is not Java 17, install it before proceeding.

Download the Application

1. Go to the project repository:
   https://github.com/AY2526S2-CS2113-W13-1/tp
2. Navigate to the **Releases** section.
3. Download the latest `.jar` file.

Running the Application

1. Open a terminal/command prompt.
2. Navigate to the folder containing the `.jar` file.
3. Run the following command:

   ```
   java -jar CCAledger.jar
   ```

First Run

- The application will start in the terminal.
- You can begin entering commands immediately.
- Use the following command to see available commands:

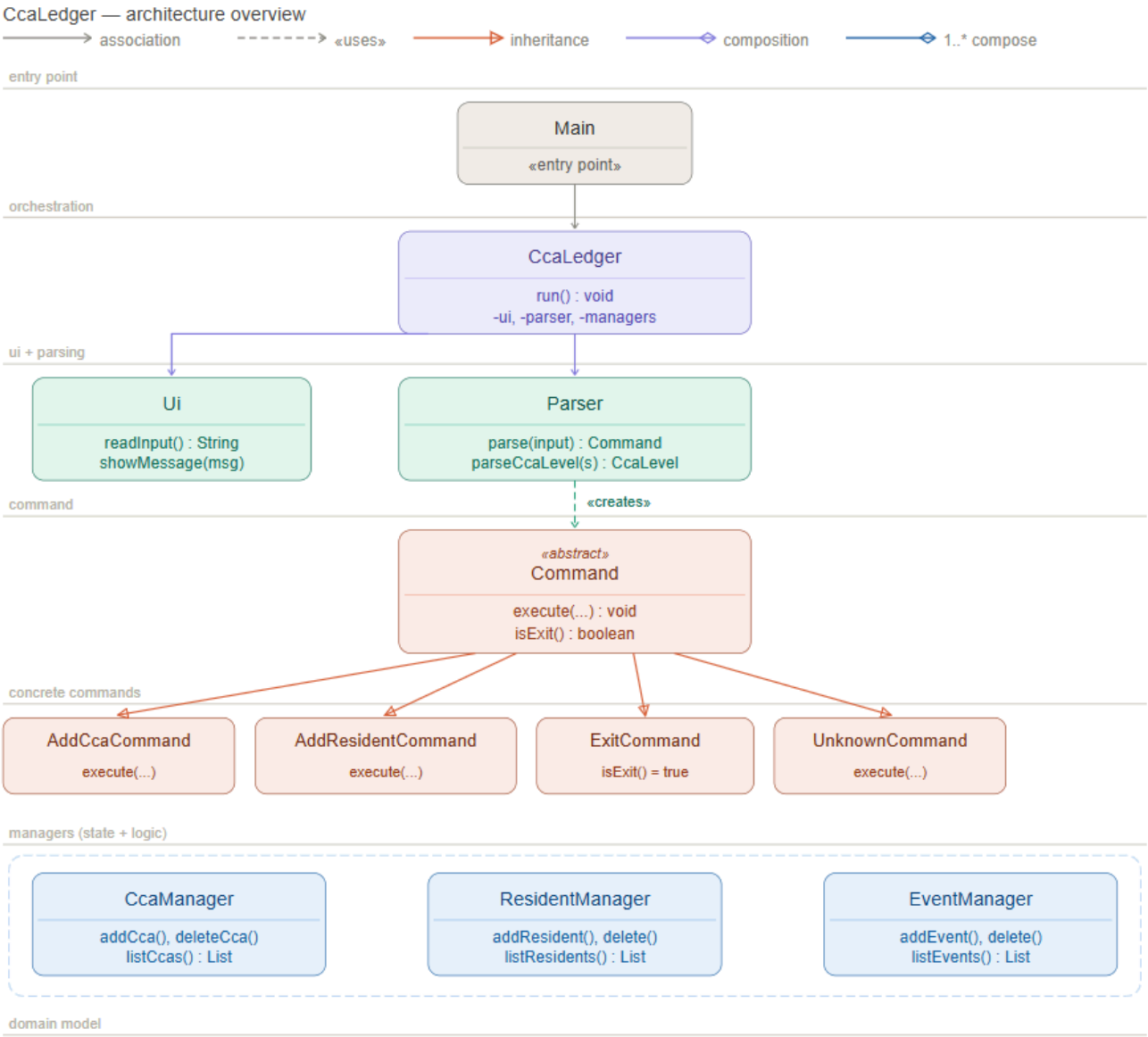
  ```
  help
  ```

Notes

- Ensure that the `.jar` file is in the correct directory before running.
- If the application does not start, re-check your Java version and installation.
- All data will be stored locally in the same directory as the `.jar` file.

Overall Architecture

CcaLedger follows a **layered, command-driven architecture**. Each layer has a single responsibility and communicates only with its immediate neighbours.



The diagram above shows the six layers and their relationships. The table below summarises each layer's role.

Layer	Key Classes	Responsibility
Entry Point	`Main`	Instantiates `CcaLedger` and calls `run()`.
Orchestration	`CcaLedger`	Owns the main loop; coordinates all components.
UI & Parsing	`Ui`, `Parser`	Handles console I/O; translates input into `Command` objects.
Command	`Command` and subclasses	Encapsulates a single user-facing operation.

Layer	Key Classes	Responsibility
Managers	`CcaManager`, `ResidentManager`, `EventManager`	Holds and mutates application state.
Domain Model	`Cca`, `Resident`, `Event`, `CcaLevel`	Plain data objects with no business logic.

How a command executes (happy path):

1. `Ui.readInput()` reads a line from the user.
2. `Parser.parse(input)` inspects the string and returns the appropriate `Command` subclass.
3. `CcaLedger` calls `command.execute(ccaManager, residentManager, eventManager, ui)`.
4. The command calls the relevant manager method(s) and prints feedback via `Ui`.
5. If `command.isExit()` returns `true`, the loop terminates.

Key design rules:

- Domain objects (`Cca`, `Resident`, `Event`) have no outward dependencies.
- Managers never reference `Ui`, `Parser`, or `Command`.
- Commands receive all dependencies as method parameters in `execute()` — they store nothing.
- `Command` subclasses are only ever instantiated inside `Parser.parse()`.

CCA Commands

Add CCA Command

Overview

The `add-cca` command adds a new CCA to the system.

Format: `add-cca <cca name> <level>`

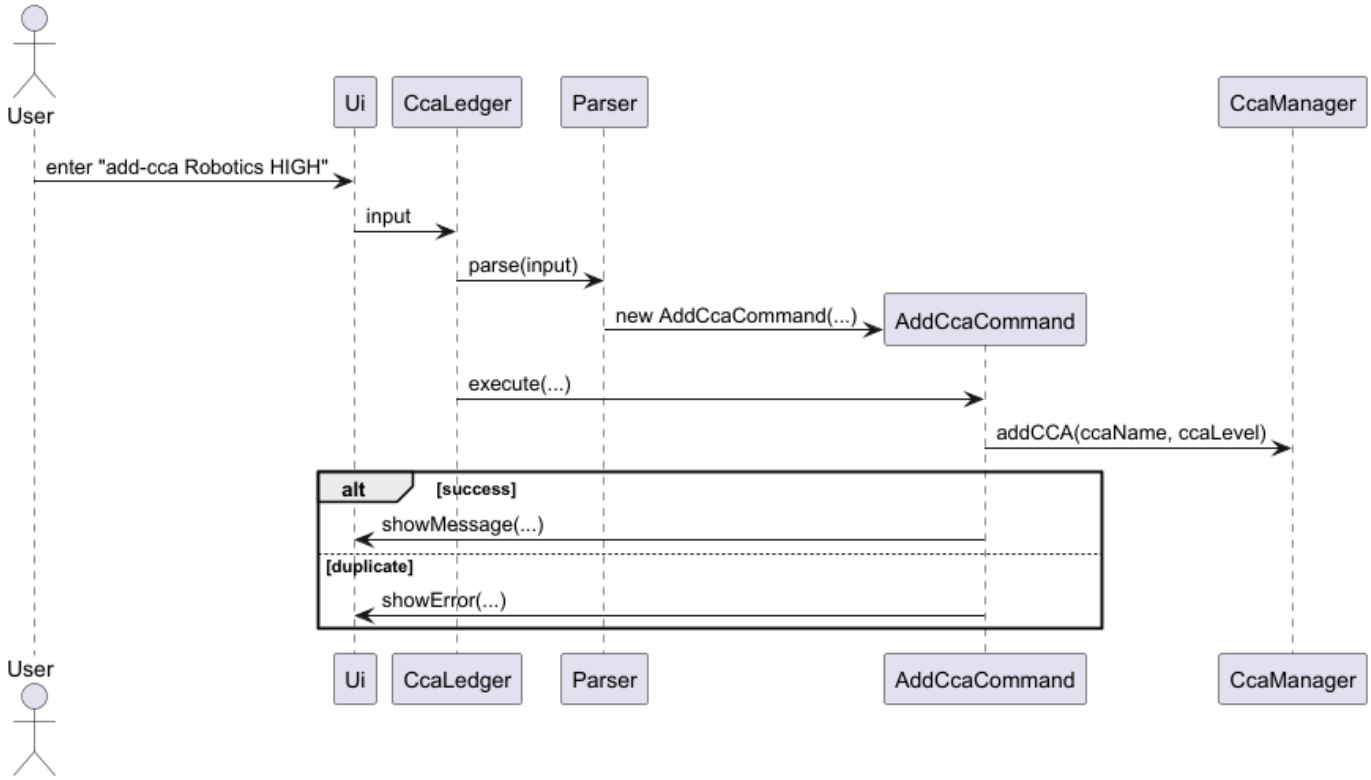
Implementation

The `add-cca` command is implemented using the Command pattern.

- The `Parser` creates an `AddCcaCommand` object from user input.
- `AddCcaCommand.execute()` calls `CcaManager.addCCA(...)`.
- If the CCA already exists, a `DuplicateCcaException` is thrown and handled.

```java
@Override
public void execute(CcaManager ccaManager, ResidentManager residentManager, EventManager e
    try {
        ccaManager.addCCA(ccaName, ccaLevel);
        ui.showMessage("CCA added: " + ccaName + "(" + ccaLevel + ")");
    } catch (DuplicateCcaException | InvalidCcaLevelException e) {
        ui.showError(e.getMessage());
    }
}
```

Sequence Diagram



Design Considerations

- Command pattern is used to separate parsing and execution.
- Exception handling is used to manage duplicate CCA cases cleanly

Alternatives Considered

1. Direct Invocation from Parser to Manager Approach: Parser directly calls `CcaManager.addCCA(...)` Rejected because:
    - Violates separation of concerns
    - Makes Parser overly complex
    - Reduces extensibility

View CCA Command

Overview

The `view-cca` command displays the list of all CCAs stored in the system.

Format: `view-cca`

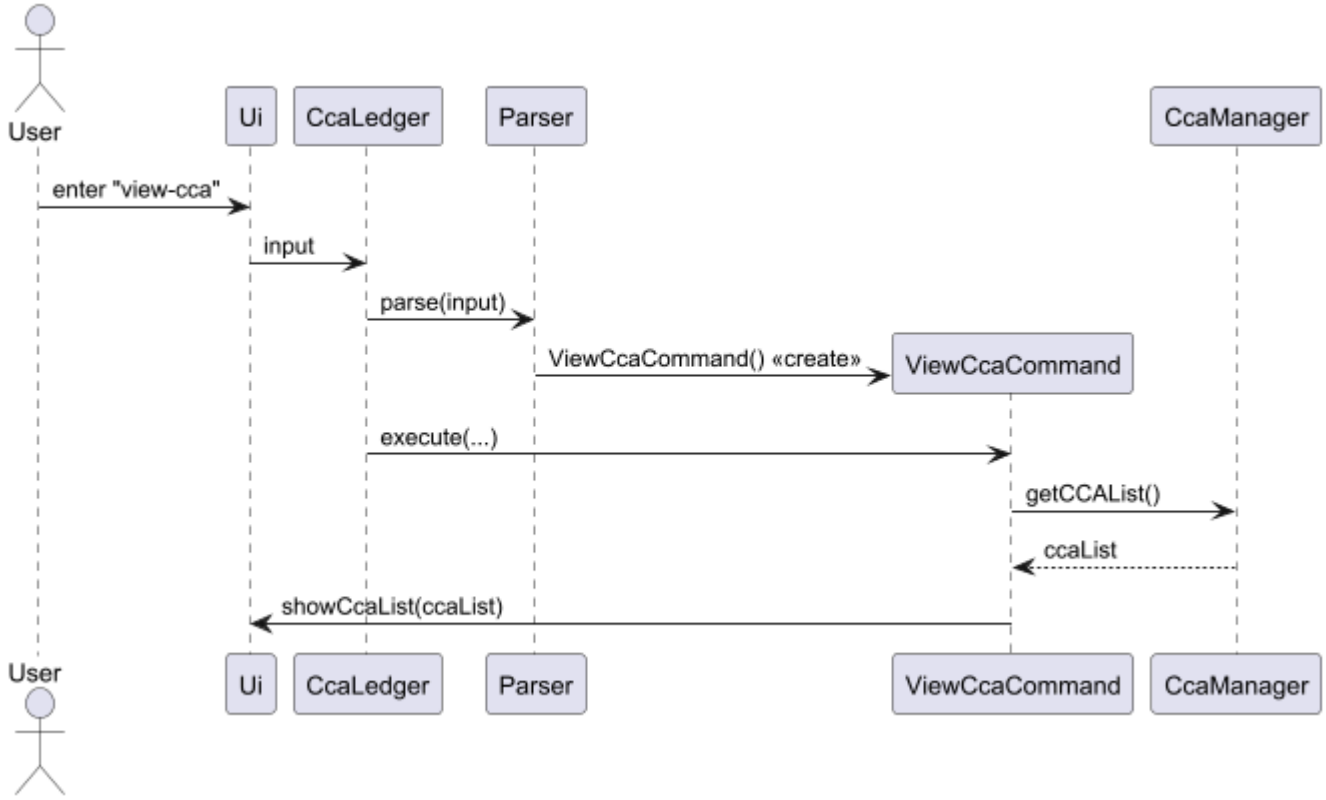
Implementation

The `view-cca` command retrieves and displays all CCAs.

- The `Parser` creates a `ViewCcaCommand` object.
- `ViewCcaCommand.execute()` calls `CcaManager.getCCAList()`.
- The retrieved list is passed to `Ui.showCcaList(...)` for display.

```
@Override
public void execute(CcaManager ccaManager, ResidentManager residentManager, Ui ui) {
    ArrayList<Cca> ccaList = ccaManager.getCCAList();
    ui.showCcaList(ccaList);
}
```

Sequence Diagram



Delete CCA Command

Overview

The `delete-cca` command removes an existing CCA from the system.

Format: `delete-cca <cca name>`

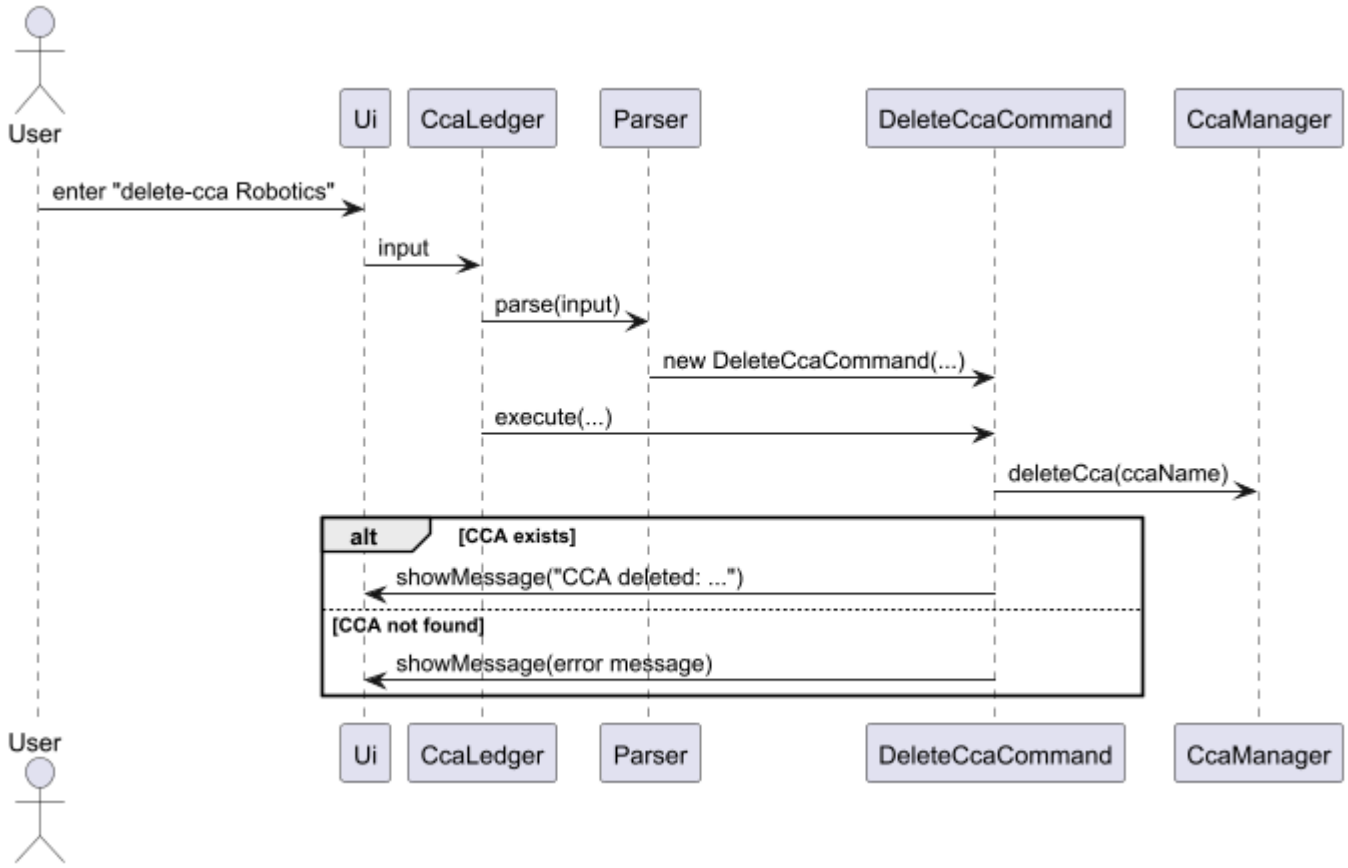
Implementation

The `delete-cca` command is implemented using the Command pattern.

- The `Parser` creates a `DeleteCcaCommand` object from user input.
- `DeleteCcaCommand.execute()` calls `CcaManager.deleteCca(...)`.
- If the CCA does not exist, a `CcaNotFoundException` is thrown and handled.

```java
@Override
public void execute(CcaManager ccaManager, ResidentManager residentManager, Ui ui) {
  try {
      ccaManager.deleteCca(ccaName);
      ui.showMessage("CCA deleted: " + ccaName);
  } catch (CcaNotFoundException e) {
      ui.showMessage(e.getMessage());
  }
}
```

Sequence Diagram



Design Considerations

- Command pattern is used to separate parsing and execution.
- Exception handling is used to manage non-existent CCA cases cleanly.

Alternatives Considered

1. Direct Invocation from Parser to Manager
   Approach: Parser directly calls `CcaManager.deleteCca(...)`
   Rejected because:
   - Violates separation of concerns
   - Makes Parser overly complex
   - Reduces extensibility

Add EXCO to CCA Command

Overview

The `add-exco-to-cca` command adds an existing resident as an EXCO for the Cca.

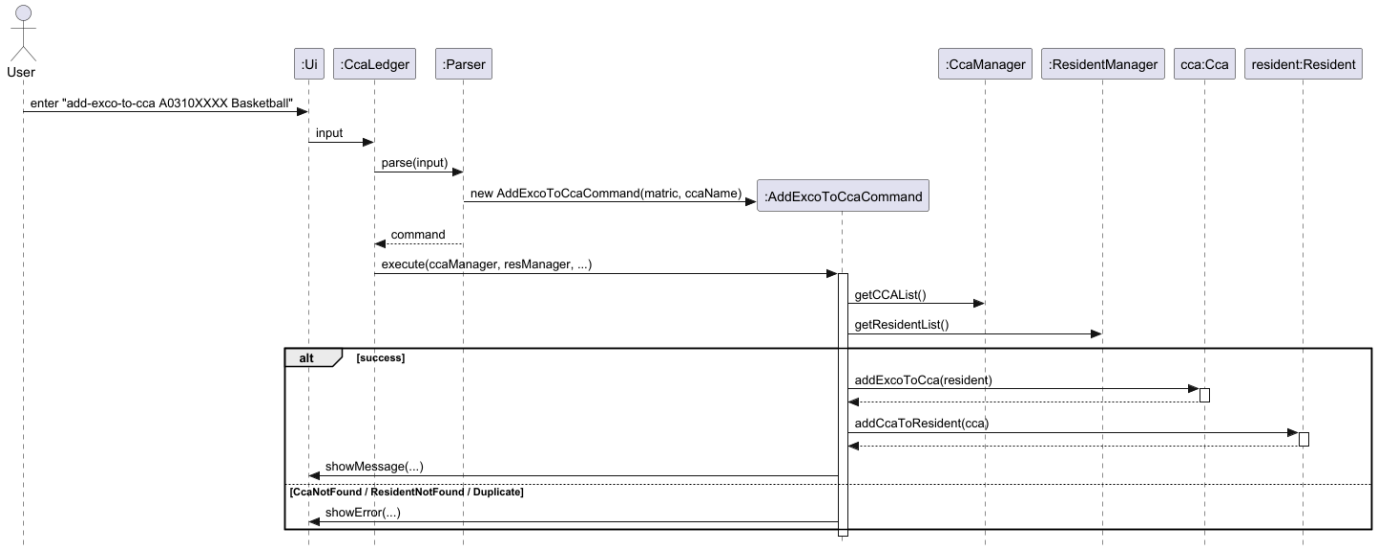
Format: `add-exco-to-cca <matric number> <cca name>`

Implementation

- The `Parser` creates a `AddExcoToCcaCommand` object.
- The command retrieves the `Resident` from `ResidentManager`.
- The corresponding `Cca` is retrieved from `CcaManager`.
- The `Resident` is added to the Cca as an EXCO in the `excoResidents` arraylist.
- Exceptions are thrown if the resident or CCA does not exist.

```
@Override
public void execute(CcaManager ccaManager, ResidentManager residentManager, EventManager e
   try {
      Cca cca = ccaManager.getCCAList().stream()
              .filter(x -> x.getName().equals(ccaName))
              .findFirst()
              .orElseThrow(() -> new CcaNotFoundException(ccaName + " not found."));

      Resident resident = residentManager.getResidentList().stream()
              .filter(x -> x.getMatricNumber().equals(matriculationNo))
              .findFirst()
              .orElseThrow(() -> new ResidentNotFoundException(matriculationNo + " not fou

      cca.addExcoToCca(resident);
      resident.addCcaToResident(cca);

      ui.showMessage("Resident " + resident + " was added as an EXCO to CCA: " + cca.getNa

   } catch (CcaNotFoundException | ResidentNotFoundException | ResidentAlreadyInCcaExcepti
      ui.showError(e.getMessage());
   }
}
```

Sequence Diagram



View all the EXCOs of a CCA

Overview

The `view-exco` command display the list of EXCOs of an existing Cca.

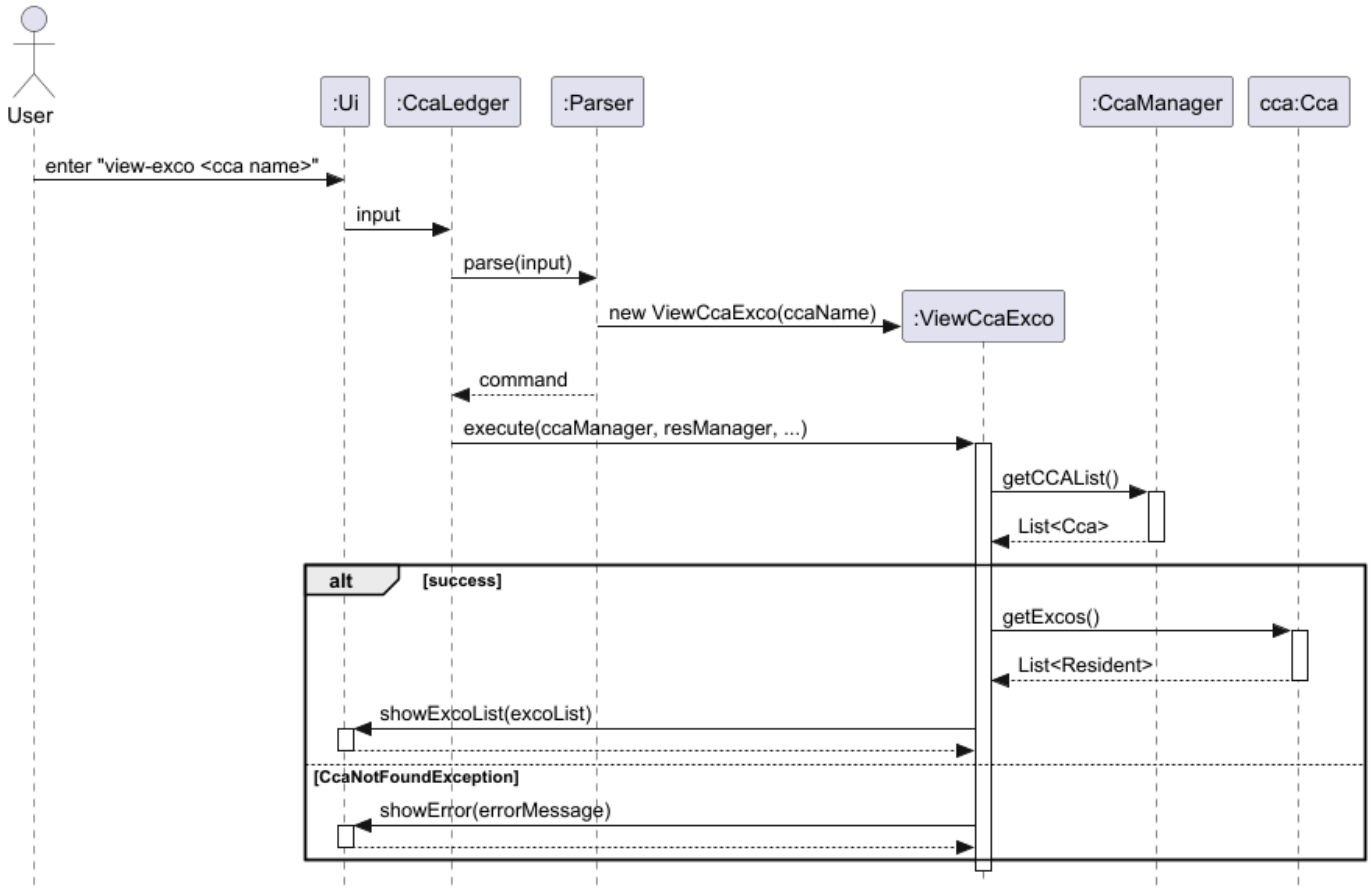
Format: `view-exco <cca name>`

Implementation

- The `Parser` creates a `ViewCcaExco` object.
- The command retrieves the `CcaList` from `CcaManager`.
- It checks if the `Cca` input is a part of `CcaList`
- If no it throws a `CcaNotFoundException`.
- If yes, then is displays the `excoMembers` of a `Cca`

```
@Override
public void execute(CcaManager ccaManager, ResidentManager residentManager, EventManager e
   try {
      Cca cca = ccaManager.getCCAList().stream()
              .filter(x -> x.getName().equals(ccaName))
              .findFirst()
              .orElseThrow(() -> new CcaNotFoundException(ccaName + " not found."));
      ui.showExcoList(cca.getExcos());
   } catch (CcaNotFoundException e) {
      ui.showError(e.getMessage());
   }
}
```

Sequence Diagram



CCA Statistics Command

Overview

The `cca-stats` command displays the average points and most active member for each CCA as well as the most popular CCA based on the average points.

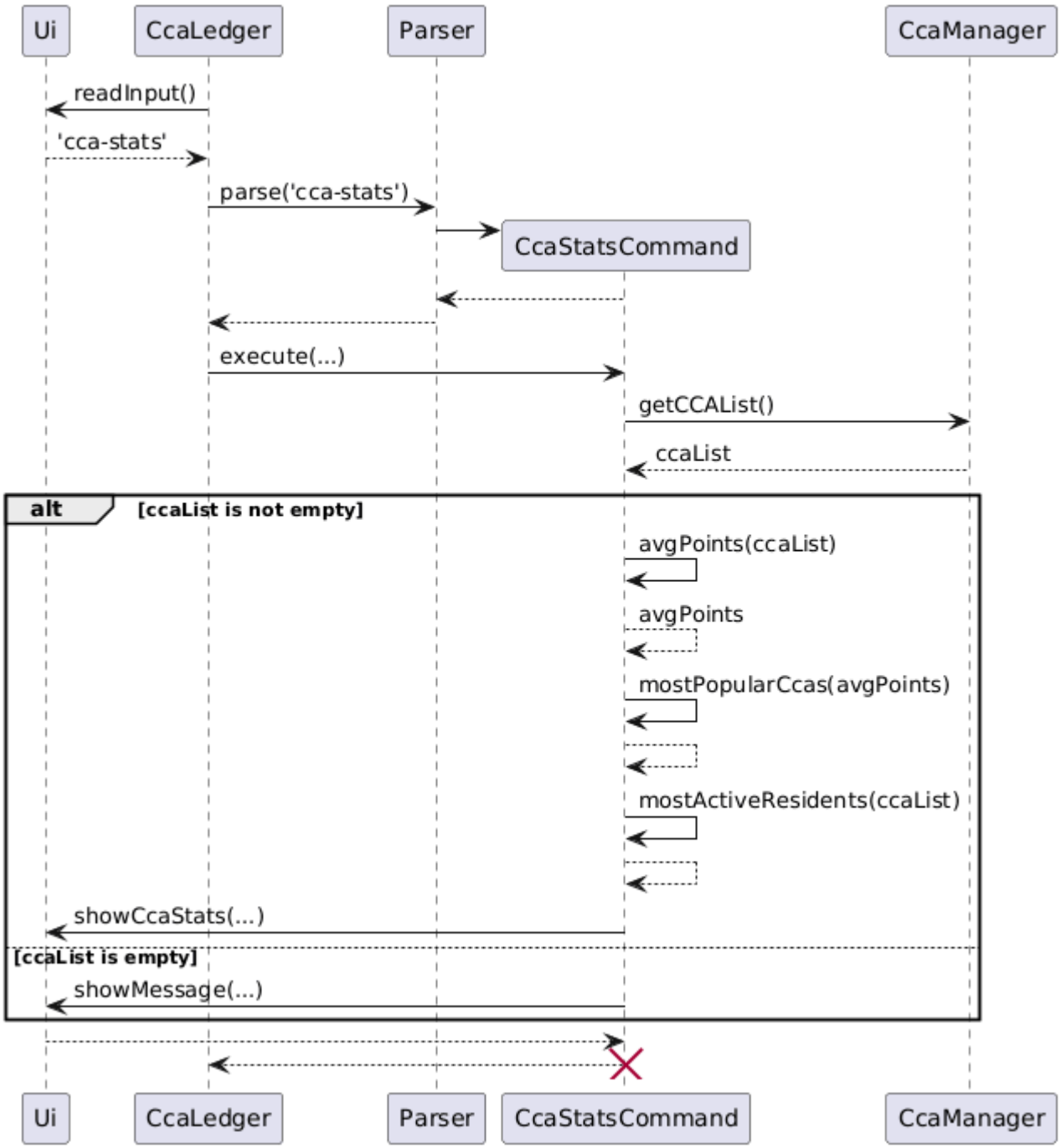
Format: `cca-stats`

Implementation

- The `Parser` creates a `CcaStatsCommand` object.
- `CcaStatsCommand.avgPoints()` computes the average points for each CCA.
- `CcaStatsCommand.mostPopularCca()` finds the most popular CCA by finding the CCA with the highest average points.
- `CcaStatsCommand.mostActiveResidents()` finds the most active member of each CCA by taking the resident with the most points for that CCA.
- If there are no CCAs in the first place, `CcaStatsCommand.execute()` passes a message to the user through `Ui.showMessage()`. Otherwise, it passes the above information to `Ui.showCcaStats()` for display.

```java
@Override
public void execute(CcaManager ccaManager, ResidentManager residentManager, EventManager e
   ArrayList<Cca> ccas = ccaManager.getCCAList();
   try {
      HashMap<Cca, Double> avgPoints = avgPoints(ccas);
      Cca mostPopularCca = mostPopularCca(avgPoints);
      HashMap<Cca, Resident> mostActiveResidents = mostActiveResidents(ccas);
      ui.showCcaStats(avgPoints, mostPopularCca, mostActiveResidents);
   } catch (IllegalArgumentException e) {
      ui.showMessage("There are no CCAs currently. Please add CCAs using add-cca command")
   }
}
```

Sequence Diagram



Resident Commands

Add Resident Command

Overview

The `add-resident` command adds a new resident to the system.

Format:
`add-resident <resident name> <matric number>`

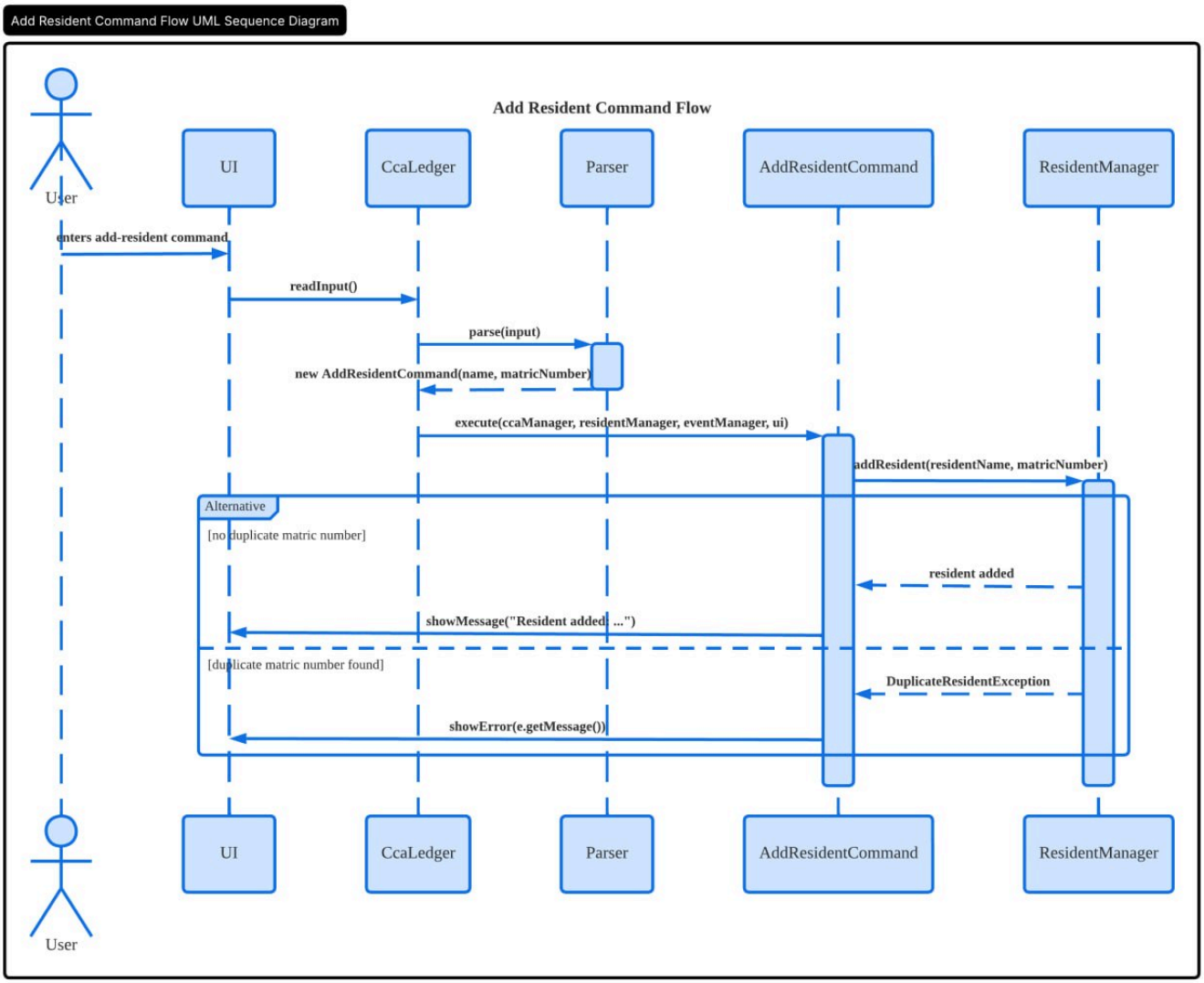
Implementation

The `add-resident` command is implemented using the Command pattern.

- The `Parser` creates an `AddResidentCommand` object from user input.
- `AddResidentCommand.execute()` calls `ResidentManager.addResident(...)`.
- If a resident with the same matric number already exists, a `DuplicateResidentException` is thrown and handled.

```
@Override
public void execute(CcaManager ccaManager, ResidentManager residentManager, EventManager e
   try {
      residentManager.addResident(residentName, matricNumber);
      ui.showMessage("Resident added: " + residentName + " " + matricNumber);
   } catch (DuplicateResidentException e) {
      ui.showError(e.getMessage());
   }
}
```

Sequence Diagram



Design Considerations

- Command pattern separates parsing and execution.
- Duplicate validation is handled inside ResidentManager, keeping business logic centralized.

View Resident Command

Overview

The `view-resident` command displays the list of all residents stored in the system.

Format: `view-resident`

Implementation

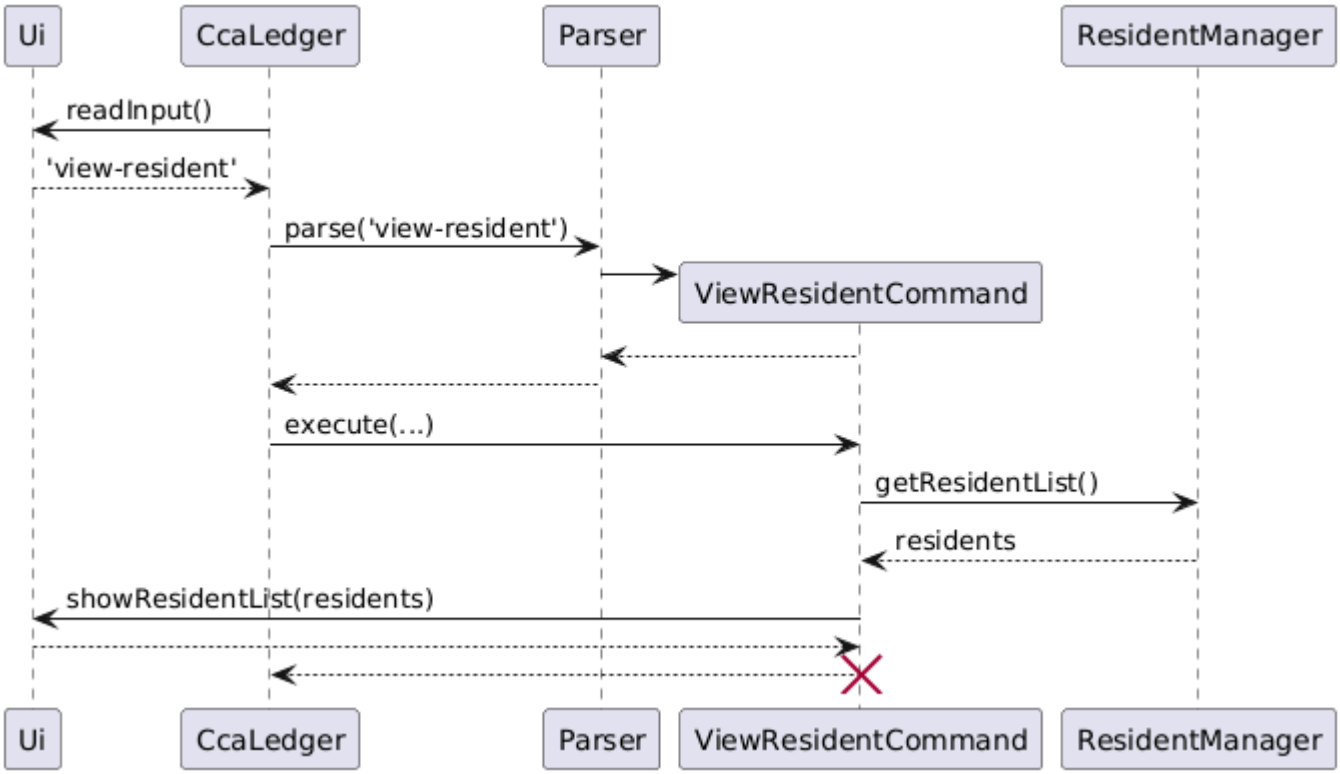
The `view-resident` command retrieves and displays all residents.

- The `Parser` creates a `ViewResidentCommand` object.

- `ViewResidentCommand.execute()` calls `ResidentManager.getResidentList()`.
- The retrieved list is passed to `Ui.showResidentList(...)` for display.

```
@Override
 public void execute(CcaManager ccaManager, ResidentManager residentManager, Ui ui) {
     ArrayList<Resident> residentList = residentManager.getResidentList();
     ui.showResidentList(residentList);
 }
```

Sequence Diagram



Delete Resident Command

Overview

The `delete-resident` command removes an existing resident from the system.

Format: `delete-resident <matric number>`

Implementation

The `delete-resident` command is implemented using the Command pattern.

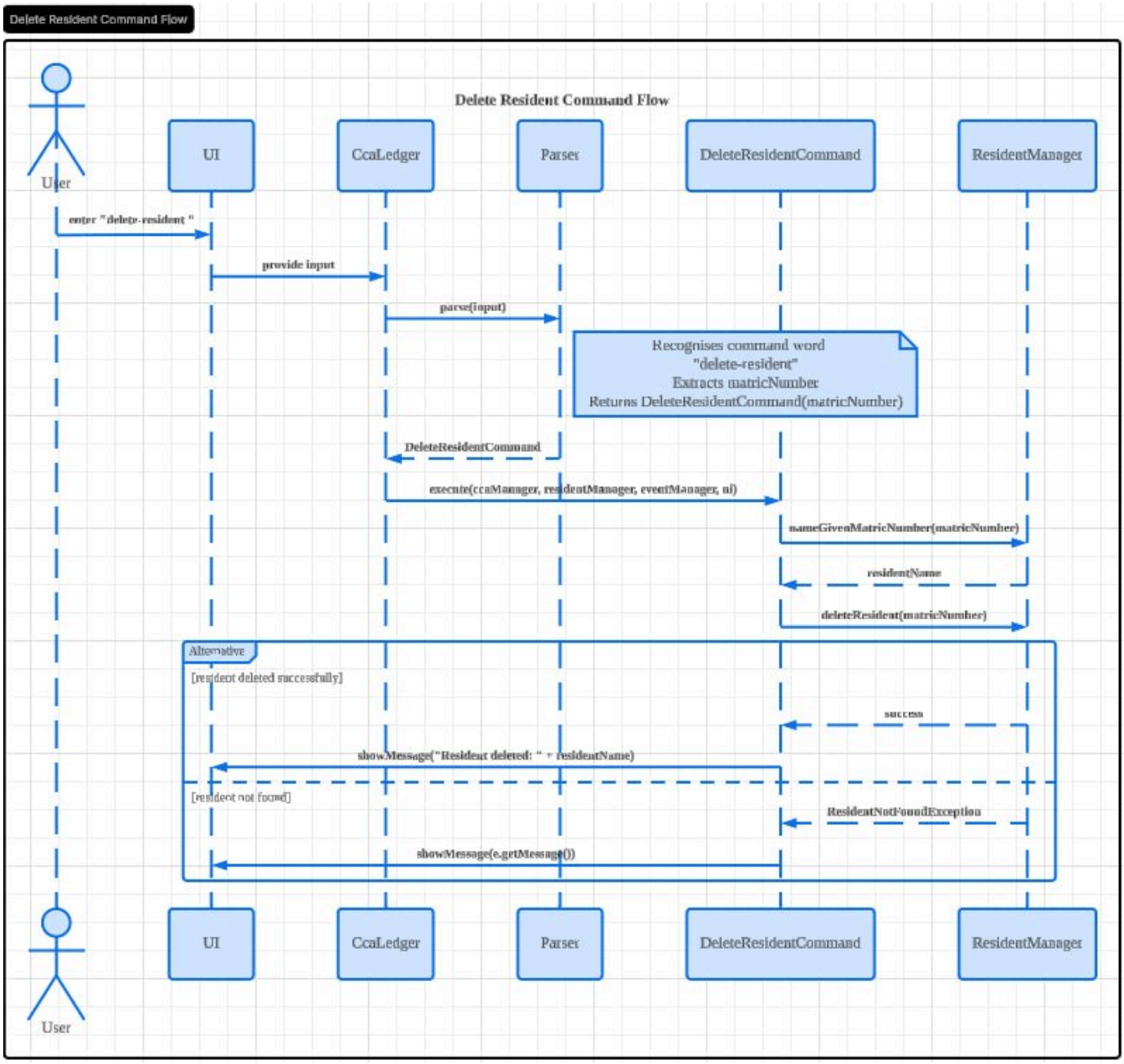
The `Parser` creates a `DeleteResidentCommand`. `DeleteResidentCommand.execute()` retrieves the resident name using `ResidentManager.nameGivenMatricNumber(...).` It then calls

`ResidentManager.deleteResident(...)`. If the resident does not exist, a ResidentNotFoundException is thrown and handled.

@Override

```
public void execute(CcaManager ccaManager, ResidentManager residentManager, EventManager e
try {
String residentName = residentManager.nameGivenMatricNumber(matricNumber);
residentManager.deleteResident(matricNumber);
ui.showMessage("Resident deleted: " + residentName);
} catch (ResidentNotFoundException e) {
ui.showMessage(e.getMessage());
}
}
```

Sequence Diagram



Design Considerations

- Delegates deletion logic fully to ResidentManager.
- Retrieves resident name before deletion for better user feedback.

Add Resident to CCA Command

Overview

The `add-resident-to-cca` command adds an existing resident to a CCA.

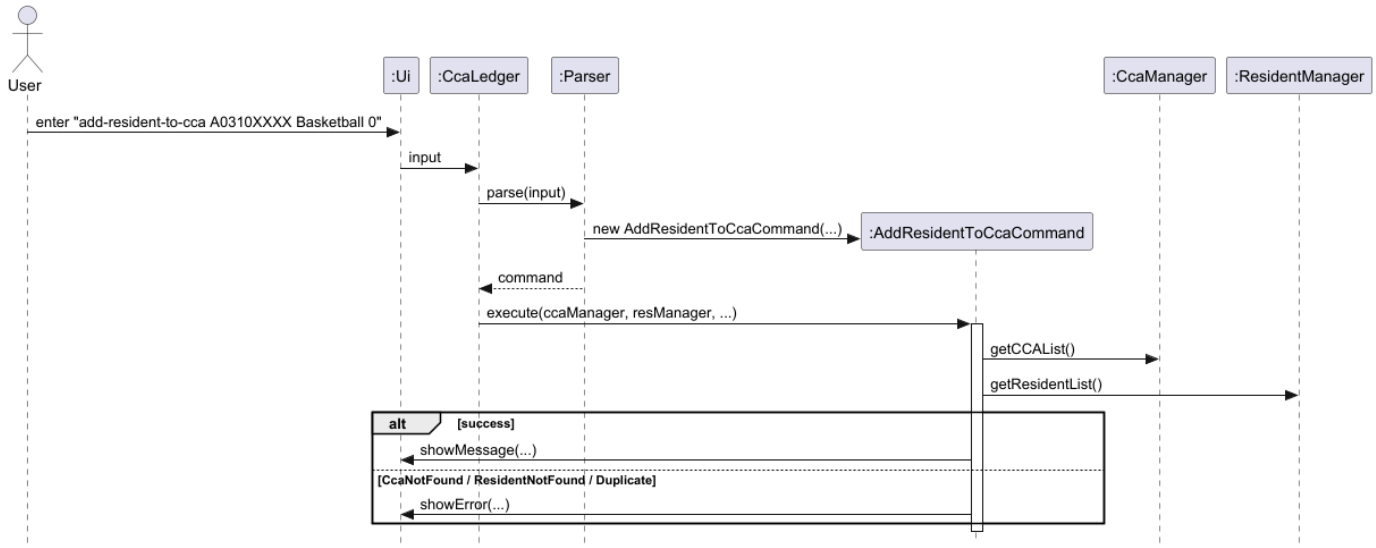
Format: `add-resident-to-cca <matric number> <cca name> <points>`

Implementation

- The `Parser` creates a `AddResdientToCcaCommmand` object.
- The command retrieves the `Resident` from `ResidentManager`.
- The corresponding `Cca` is retrieved from `CcaManager`.
- The `Resident` is added to the Cca using `Cca.addResidentToCca(...)`.
- Exceptions are thrown if the resident or CCA does not exist.

```
@Override
 public void execute(CcaManager ccaManager, ResidentManager residentManager, EventManager
     try {
         Cca cca = ccaManager.getCCAList().stream()
                 .filter(x -> x.getName().equals(ccaName))
                 .findFirst()
                 .orElseThrow(() -> new CcaNotFoundException(ccaName + " not found."));

         Resident resident = residentManager.getResidentList().stream()
                 .filter(x -> x.getMatricNumber().equals(matriculationNo))
                 .findFirst()
                 .orElseThrow(() -> new ResidentNotFoundException(matriculationNo + " not

         cca.addResidentToCca(resident);
         resident.addCcaToResident(cca, pointsScored);

         ui.showMessage("Resident " + resident + " was added to CCA: " + cca.getName() +
                 " with " + pointsScored + " points.");

     } catch (CcaNotFoundException | ResidentNotFoundException | ResidentAlreadyInCcaExcep
         ui.showError(e.getMessage());
     }
 }
```

Sequence Diagram



View Points Command

Overview

The view-points command displays the CCA points for all residents in the system.

Format: view-points

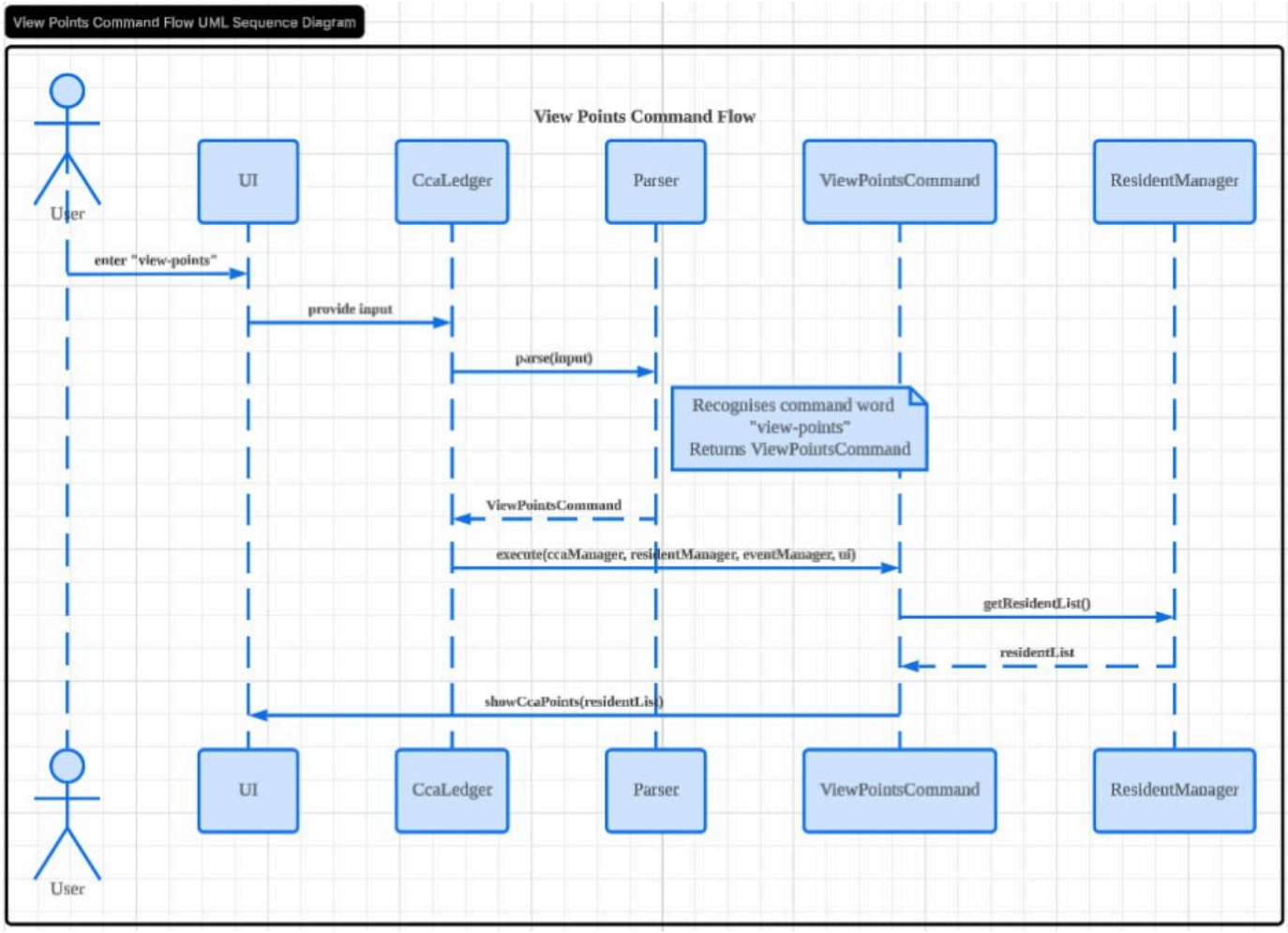
Implementation

The view-points command retrieves and displays CCA points for all residents.

- The Parser creates a ViewPointsCommand object.
- ViewPointsCommand.execute() calls ResidentManager.getResidentList().
- The retrieved list is passed to Ui.showCcaPoints(...) for display.

```
@Override
public void execute(CcaManager ccaManager, ResidentManager residentManager, Ui ui) {
ArrayList<Resident> residentList = residentManager.getResidentList();
ui.showCcaPoints(residentList);
}
```

Sequence Diagram



Design Considerations

- Command pattern is used to separate parsing and execution.
- Reuses ResidentManager.getResidentList() to retrieve resident data, keeping the manager layer lean.

Resident Statistics Command

Overview

The `resident-stats` command displays the total points for each resident and the most active residents across all CCAs

Format: `resident-stats`

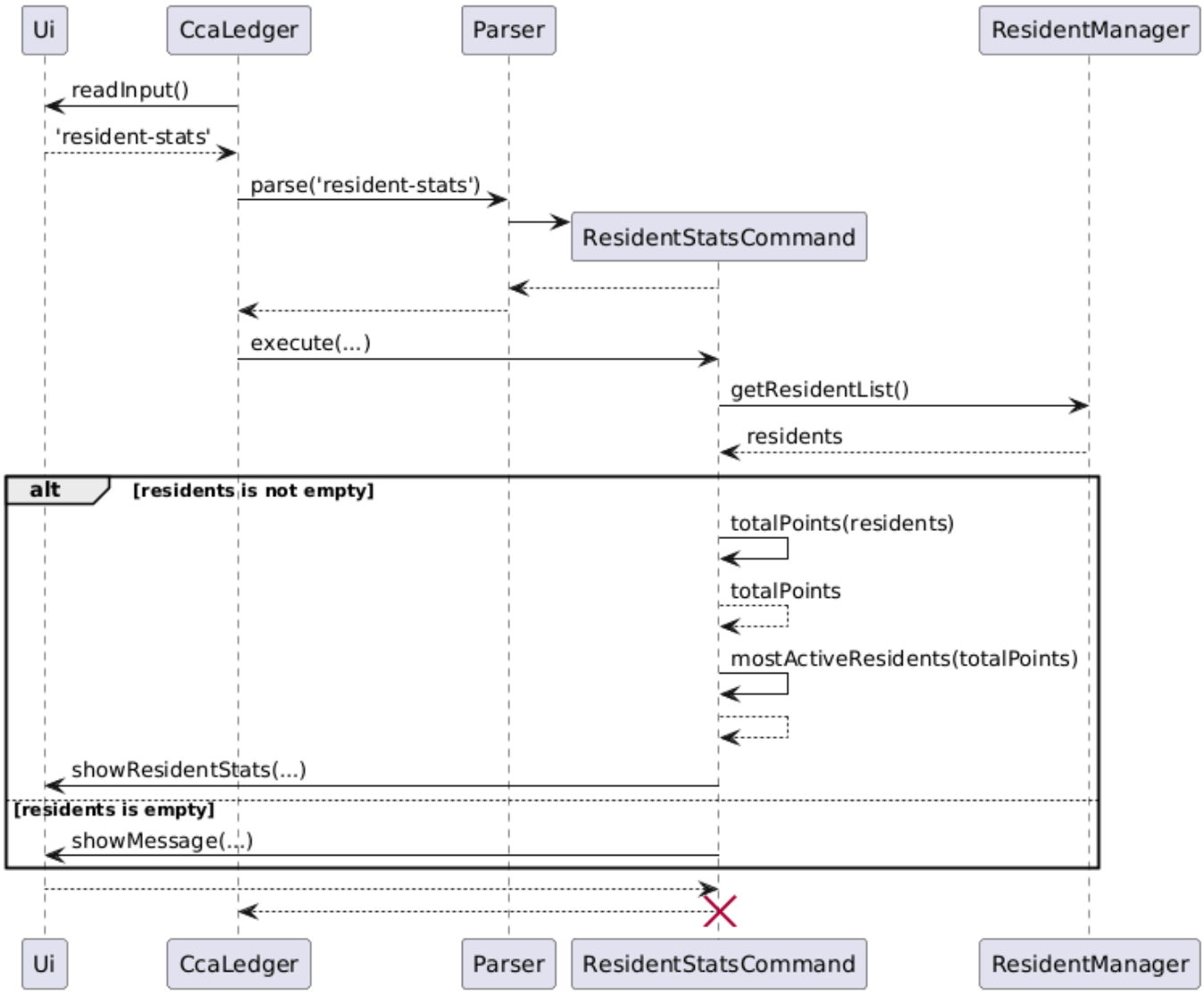
Implementation

- The `Parser` creates a `ResidentStatsCommand` object.
- `ResidentStatsCommand.totalPoints()` computes the total points for each resident.

- `ResdientStatsCommand.mostActiveResidents()` finds the most active residents across all CCAs based on their total points.
- If there are no resdients in the first place, `ResidentStatsCommand.execute()` passes a message to the user through `Ui.showMessage()`. Otherwise, it passes the above information to `Ui.showResidentStats()` for display.

```
@Override
public void execute(CcaManager ccaManager, ResidentManager residentManager, EventManager e
    ArrayList<Resident> residents = residentManager.getResidentList();
    if (residents.isEmpty()) {
        ui.showMessage("There are no residents currently. Please add residents using add-res
        return;
    }
    HashMap<Resident, Integer> totalPoints = totalPoints(residents);
    ArrayList<Resident> mostActiveResident = mostActiveResidents(totalPoints);
    ui.showResidentStats(totalPoints, mostActiveResident);
}
```

Sequence Diagram



Event Commands

Add Event Command

Overview

The `add-event` command adds a new event under a specified CCA.

Format: `add-event <event name> <cca name> <date/time>`

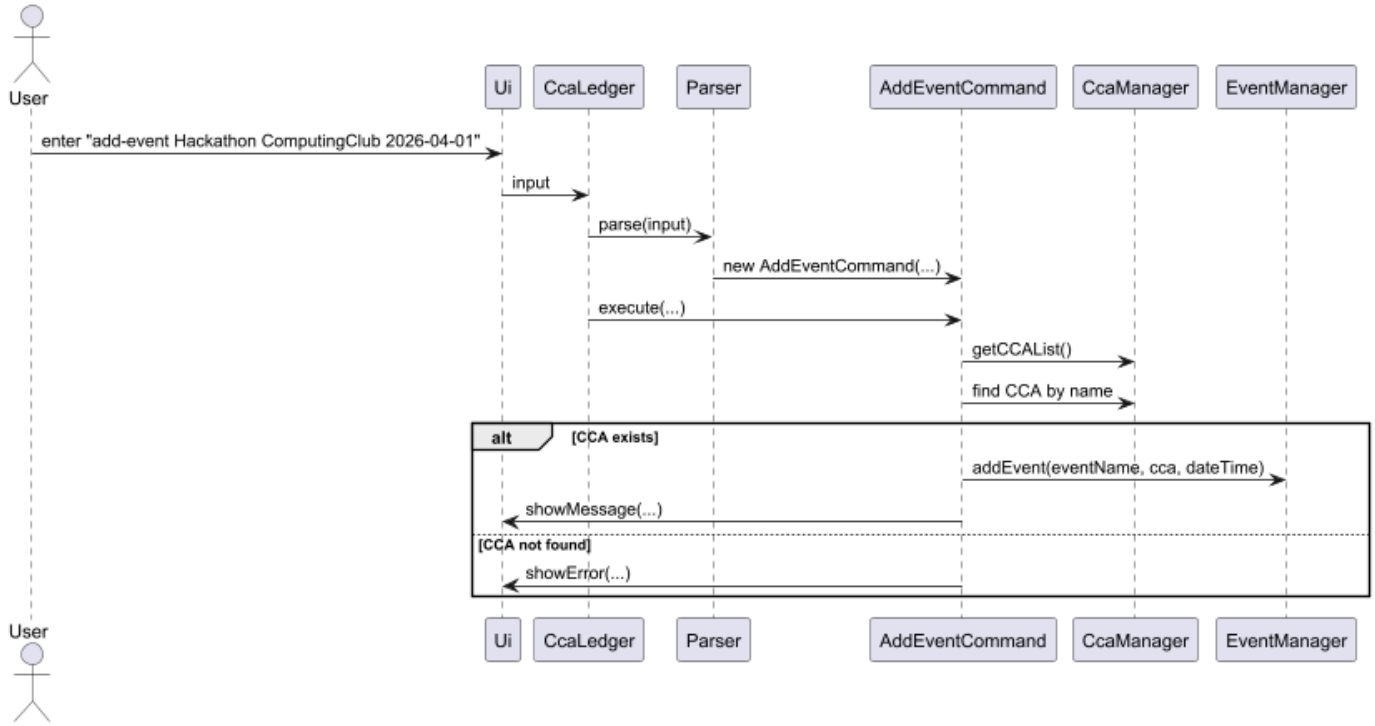
Implementation

The `add-event` command is implemented using the Command pattern.

- The `Parser` creates an `AddEventCommand` object from user input.
- `AddEventCommand.execute()` retrieves the corresponding CCA from `CcaManager`.
- The event is added using `EventManager.addEvent(...)`.
- If the CCA does not exist, a `CcaNotFoundException` is thrown and handled.

```
@Override
public void execute(CcaManager ccaManager, ResidentManager residentManager, EventManager e
   try {
      Cca cca = ccaManager.getCCAList().stream()
              .filter(x -> x.getName().equals(ccaName))
              .findFirst()
              .orElseThrow(() -> new CcaNotFoundException(ccaName + " not found."));

      eventManager.addEvent(eventName, cca, dateTime);

      ui.showMessage("Event added: " + eventName + " for the CCA " + ccaName + ", during "

   } catch (CcaNotFoundException e) {
      ui.showError(e.getMessage());
   }
}
```

Sequence Diagram



Add Resident to Event Command

Overview

The `add-resident-to-event` command adds an existing resident to a specific event under a CCA.

Format: `add-resident-to-event <matric number> <event name> <cca name>`

Implementation

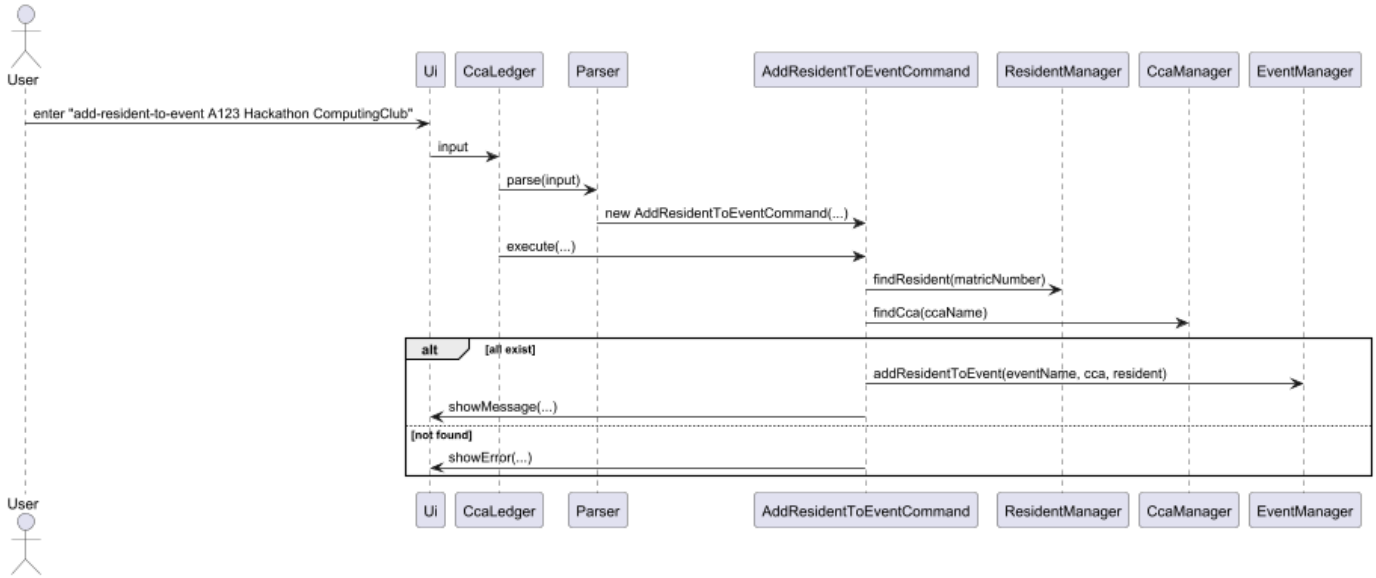
The command is implemented using the Command pattern.

- The `Parser` creates an `AddResidentToEventCommand`.
- The command retrieves the `Resident` from `ResidentManager`.
- The corresponding `Cca` is retrieved from `CcaManager`.
- The resident is added to the event using `EventManager.addResidentToEvent(...)`.
- Exceptions are thrown if the resident, CCA, or event does not exist.

```
@Override
public void execute(CcaManager ccaManager, ResidentManager residentManager, EventManager e
    try {
        Resident resident = residentManager.getResidentList().stream()
                .filter(r -> r.getMatricNumber().equalsIgnoreCase(matricNumber))
                .findFirst()
                .orElseThrow(() -> new ResidentNotFoundException(...));

        Cca cca = ccaManager.getCCAList().stream()
```

```
                .filter(c -> c.getName().equalsIgnoreCase(ccaName))
                .findFirst()
                .orElseThrow(() -> new CcaNotFoundException(...));

        eventManager.addResidentToEvent(eventName, cca, resident);

        ui.showMessage(...);

    } catch (ResidentNotFoundException | CcaNotFoundException | EventNotFoundException e) {
        ui.showError(e.getMessage());
    }
}
```

Sequence Diagram



View My Events Command

Overview

The `view-my-event` command displays all events that a resident is participating in.

Format:
`view-my-event <matric number>`

Implementation

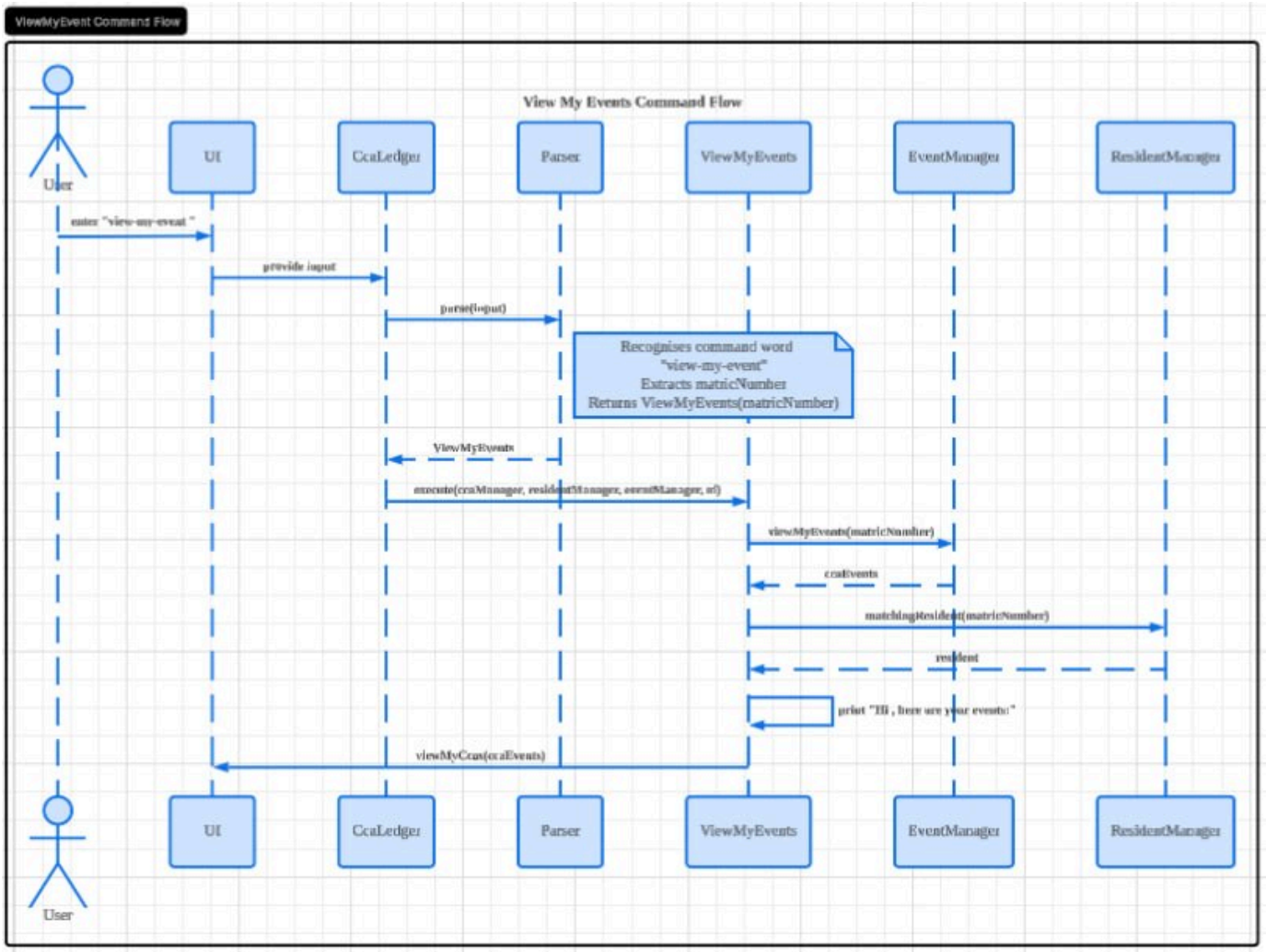
The `view-my-event` command retrieves and displays all events associated with a resident.

The `Parser` creates a `ViewMyEvents` object from user input. `ViewMyEvents.execute()` calls `EventManager.viewMyEvents(matricNumber)` to retrieve the matching events. It then retrieves the

resident using `ResidentManager.matchingResident(...)`, prints a greeting using the resident's name, and passes the event list to `Ui.viewMyCcas(...)` for display.

```
@Override
public void execute(CcaManager ccaManager, ResidentManager residentManager, EventManager e
    ArrayList<Event> ccaEvents = eventManager.viewMyEvents(matricNumber);
    Resident resident = residentManager.matchingResident(matricNumber);
    System.out.println("Hi " + resident.getName() + ", here are your events: ");
    ui.viewMyCcas(ccaEvents);
}
```

Sequence Diagram



View CCA Events Command

Overview

The `view-cca-event` command displays all events under a specified CCA.

Format:
`view-cca-event <cca name>`

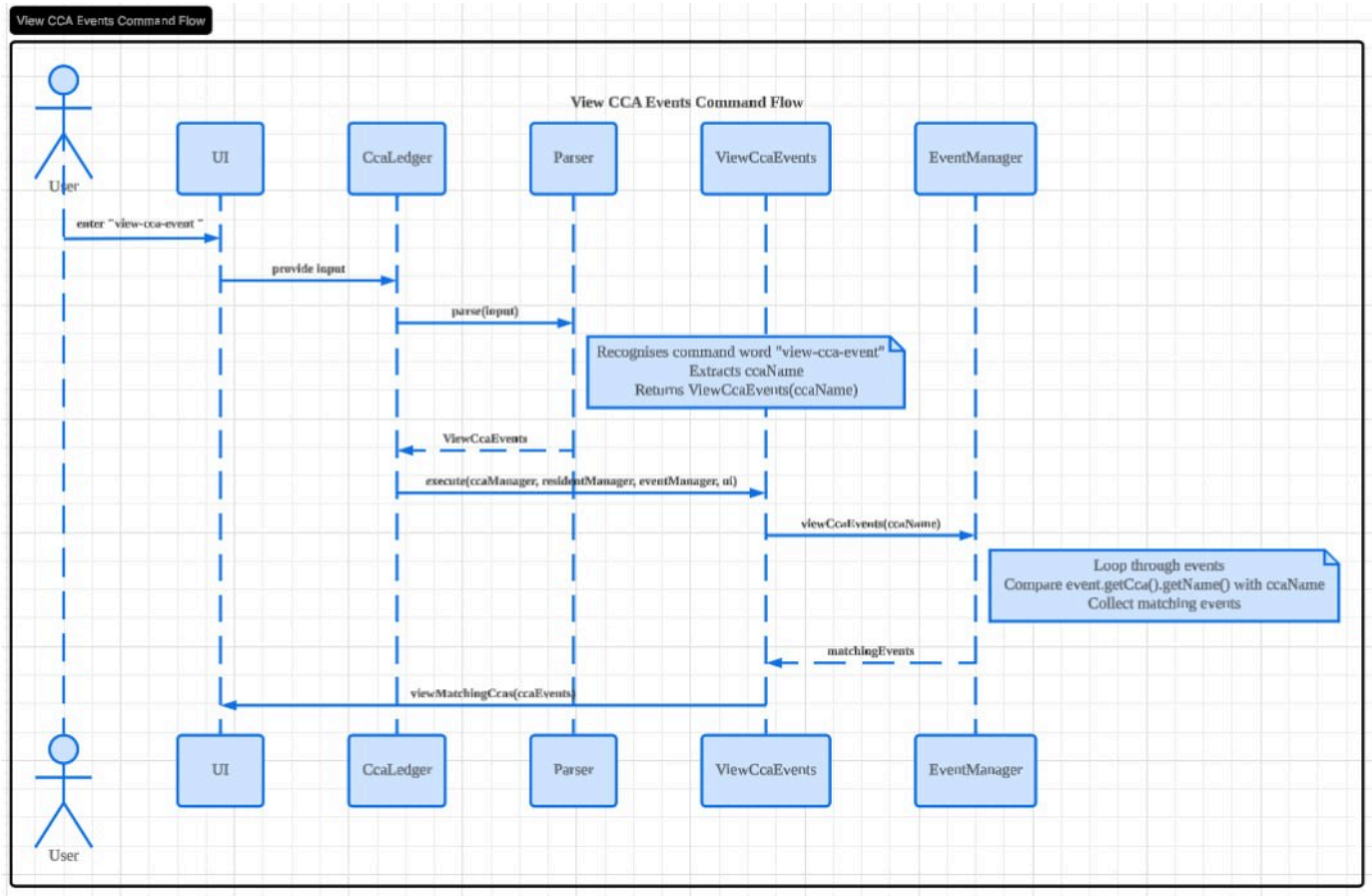
Implementation

The `view-cca-event` command retrieves and displays all events belonging to a specific CCA.

The `Parser` creates a `ViewCcaEvents` object from user input. `ViewCcaEvents.execute()` calls `EventManager.viewCcaEvents(ccaName)` to retrieve the matching events, and the resulting list is passed to `Ui.viewMatchingCcas(...)` for display.

```
@Override
public void execute(CcaManager ccaManager, ResidentManager residentManager, EventManager e
    ArrayList<Event> ccaEvents = eventManager.viewCcaEvents(ccaName);
    ui.viewMatchingCcas(ccaEvents);
}
```

Sequence Diagram



General Commands

Help Command

Overview

The `help` command presents a list of all available commands and their usage.

Format: `help`

Implementation

The `help` command is implemented using the Command pattern.

- The `Parser` creates a `HelpCommand` object from user input.
- `HelpCommand.execute()` creates a string which is a list of all commands and their usage.
- It then passes the string to `Ui.showMessage()`.

```
@Override
 public void execute(CcaManager ccaManager, ResidentManager residentManager, EventManager
    String help = "Here is a list of all commands:\n" +
            "> add-cca <cca name> <level (HIGH, MEDIUM, LOW or UNKNOWN)>\n" +
            "> view-cca\n" +
            "> delete-cca <cca name>\n" +
            "> add-event <event name> <cca name> <data time>\n" +
            "> add-resident <name> <matric number>\n" +
            "> view-resident\n" +
            "> add-resident-to-cca <matric number> <cca name> <points>\n" +
            "> add-resident-to-event <matric number> <event name> <cca name>\n" +
            "> view-points\n" +
            "> cca-stats\n" +
            "> resident-stats\n" +
            "> help\n" +
            "> bye";
    ui.showMessage(help);
}
```

Appendices

Appendix A: Product Scope

Target User Profile

- Hall Leaders / CCA Leaders
- Manage multiple CCAs and events

- Track resident participation and performance
- Prefer a fast, CLI-based tool over GUI applications
- Need quick access to statistics and summaries

Value Proposition

- Centralised management of CCAs, residents, and events
- Efficient tracking of participation and points
- Quick retrieval of statistics for decision making
- Lightweight and fast CLI interface

Appendix B: User Stories

Priority	As a ...	I want to ...	So that I can ...
High	Hall Leader	add a CCA	manage different clubs
High	Hall Leader	delete a CCA	remove inactive clubs
High	Hall Leader	view CCAs	see all available clubs
High	Hall Leader	add a resident	keep track of participants
High	Hall Leader	delete a resident	remove inactive members
High	Hall Leader	view residents	see all participants
High	Hall Leader	add an event	organize activities
High	Hall Leader	add residents to events	track participation
High	Hall Leader	view events under a CCA	monitor activities
Medium	Hall Leader	assign EXCO roles	manage leadership
Medium	Hall Leader	view EXCO members	track leadership structure
Medium	Hall Leader	view points	track performance
Medium	Hall Leader	view statistics	analyze engagement
Low	Hall Leader	view my events	track individual participation

---

Appendix C: Non-Functional Requirements

1. **Usability**

    - The system should be easy to use via CLI commands.
    - Commands should follow a consistent format.
2. **Performance**
    - The application should respond within 1 second for typical commands.
3. **Reliability**
    - Data should persist between sessions.
    - The system should handle invalid inputs gracefully.
4. **Portability**
    - The application should run on any system with Java 17 or above.
5. **Maintainability**
    - Code should be modular and easy to extend.
6. **Scalability**
    - The system should handle increasing numbers of residents, CCAs, and events without significant slowdown.

Appendix D: Glossary

Term	Meaning
CCA	Co-Curricular Activity
Resident	A student participating in CCAs
EXCO	Executive Committee member of a CCA
Matric Number	Unique identifier for a resident
Event	Activity organized under a CCA
Points	Score assigned based on participation
CLI	Command Line Interface

Appendix E: Instructions for Manual Testing

Overview

This section provides a guided path for testers to explore the main features of the application. It complements the User Guide and focuses on key test flows.

Initial Setup

1. Download the `.jar` file form the GitHub Repo
2. Double click the `.jar` file to launch the programme.

Core Feature Testing Flow

1. Add CCAs

```
add-cca Basketball HIGH
add-cca Dance LOW
```

---

2. View CCAs

```
view-cca
```

---

3. Add Residents

```
add-resident John A1234567B
add-resident James A7654321C
```

---

4. View Residents

```
view-resident
```

---

5. Add Events

```
add-event Practice-Week1 Basketball 29/3/26
add-event Orientation Dance 2/4/26
```

---

6. Add Residents to Events

```
add-resident-to-event A1234567B Practice-Week1 Basketball
add-resident-to-event A7654321C Orientation Dance
```

---

7. View Events

```
view-cca-event Basketball
view-my-event A1234567B
```

---

8. Assign EXCO

```
add-exco-to-cca A1234567B Basketball
view-exco Basketball
```

---

9. View Points and Statistics

```
view-points
cca-stats
resident-stats
```

---

10. Delete Operations

```
delete-resident A1234567B
delete-cca Basketball
```

Edge Cases to Try

- Adding duplicate CCAs
- Adding duplicate residents
- Adding events to non-existent CCAs
- Adding residents to non-existent events

Acknowledgements

- No external code was directly reused unless otherwise stated.
- Standard Java libraries were used for implementation.